ບົດທີ 26 ສົມຕໍາລາຂອງຕໍາລາໄຕມຸມມິຕິ ແລະ ຕໍາລາປີ້ນຂອງຕໍາລາໄຕມຸມມິຕິ

ເມື່ອຮຽນຈົບບົດນີ້ນັກຮຽນຈະສາມາດຄິດໄລ່ສົມຕໍາລາຂອງຕໍາລາໄຕມຸມມິຕິ ແລະ ຕໍາລາປີ້ນຂອງຕໍາລາໄຕມຸມມິຕິ.

1. ສົມຕໍາລາຂອງຕໍາລາໄຕມຸມມິຕິ

ຊີ້ແຈງວ່າ $(\sin x)' = \cos x$

ຕາມນິຍາມເຮົາໄດ້ $(\sin x)' = \lim_{\Delta x \to 0} \dfrac{\sin(x+\Delta x) - \sin x}{\Delta x}$

$$= \lim_{\Delta x \to 0} \frac{\sin x \cos \Delta x + \cos x \sin \Delta x - \sin x}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \left(\left(\frac{\cos \Delta x - 1}{\Delta x} \right) \sin x + \left(\frac{\sin \Delta x}{\Delta x} \right) \cos x \right)$$

$$= \left(\lim_{\Delta x \to 0} \left(\frac{\cos \Delta x - 1}{\Delta x} \right) \right) \sin x + \left(\lim_{\Delta x \to 0} \left(\frac{\sin \Delta x}{\Delta x} \right) \right) \cos x$$

$$= 0 \times \sin x + 1 \times \cos x$$

$$= \cos x$$

ດັ່ງນັ້ນ, $(\sin x)' = \cos x$

ຄໍາຖາມ 1: ຈົ່ງຊີ້ແຈງວ່າ $(\cos x)' = -\sin x$

ສູດຄິດໄລ່: $(\sin x)' = \cos x$

$$(\cos x)' = -\sin x$$

$$(\tan x)' = \frac{1}{\cos^2 x}$$

$$(\cot x)' = \frac{-1}{\sin^2 x}$$

ເມື່ອ u ເປັນຕໍາລາຕາມ x ເຮົາໄດ້: $(\sin u)' = u' \cos u$

$$(\cos u)' = -u' \sin u$$

$$(\tan u)' = \frac{u'}{\cos^2 u}$$

$$(\cot u)' = \frac{-u'}{\sin^2 u}$$

ຕົວຢ່າງ 1: ຊອກຫາຜົນຕໍ່ລາ

1) $f(x)=\sin\left(2x^2+1\right)$

$$f'(x)=\left(2x^2+1\right)'\cos\left(2x^2+1\right)$$
$$=4x\cos\left(2x^2+1\right)$$

2) $f(x)=\tan^3 2x$

$$f'(x)=\left(3\tan^2 2x\right)\left(\tan 2x\right)'$$
$$=\left(3\tan^2 2x\right)\frac{2}{\cos^2 2x}$$
$$=\frac{6\tan^2 2x}{\cos^2 2x}$$

3) $f(x)=\sin x-\cos 2x$

$$f'(x)=\cos x+2\sin 2x$$

4) $f(x)=\cos^4 x+\tan x^3$

$$f'(x)=\left(4\cos^3 x\right)\left(\cos x\right)'+\frac{\left(x^3\right)'}{\cos^2 x^3}$$
$$=-4\sin x\cos^3 x+\frac{3x^2}{\cos^2 x^3}$$

5) $f(x)=\sin x\cot 2x$

$$f'(x)=\left(\sin x\right)'\cot 2x+\left(\sin x\right)\left(\cot 2x\right)'$$
$$=\cos x\cot 2x+\frac{2\sin x}{\sin^2 2x}$$

ຄຳຖາມ 2: ຈົ່ງຄິດໄລ່ຜົນຕໍ່ລາຂອງຕໍ່ລາລຸ່ມນີ້

1) $f(x)=\cos\left(2x^3-5\right)$
2) $f(x)=\cot^2 3x$
3) $f(x)=\cos 2x+\sin 3x$
4) $f(x)=\sin^3 x+\cot x^2$
5) $f(x)=\cos 2x\tan 5x$

2. ຜົນຕໍ່ລາຂອງຕໍ່ລາປົ້ນຂອງຕໍ່ລາໄຕມຸມມິຕິ

ໃຫ້ $\sin y = x$ ຫຼື $y = \arcsin x$, $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$

ຄິດໄລ່ $\left(\sin^{-1} x\right)'$ ຫຼື $\left(\arcsin x\right)'$

ເອົາຜົນຕໍ່ລາເຂົ້າໄດ້: $\cos y \cdot \frac{dy}{dx} = 1$

$$\Rightarrow \frac{dy}{dx} = \left(\arcsin x\right)'$$

$$= \frac{1}{\cos y}$$

$$= \frac{1}{\sqrt{1-\sin^2 y}}$$

$$= \frac{1}{\sqrt{1-x^2}}$$

ຄຳຖາມ 3: ໃຫ້ $\cos y = x$ ຫຼື $y = \arccos x$, $-1 < x < 1$. ຈົ່ງຄິດໄລ່ $\left(\cos^{-1} x\right)'$ ຫຼື $\left(\arccos x\right)'$

ສູດຄິດໄລ່: $\left(\sin^{-1} x\right)' = \frac{1}{\sqrt{1-x^2}}$

$$\left(\cos^{-1} x\right)' = -\frac{1}{\sqrt{1-x^2}}$$

$$\left(\tan^{-1} x\right)' = \frac{1}{1+x^2}$$

$$\left(\cot^{-1} x\right)' = -\frac{1}{1+x^2}$$

ເມື່ອ u ເປັນຕໍ່ລາຕາມ x ເຮົາໄດ້: $\left(\sin^{-1} u\right)' = \frac{u'}{\sqrt{1-u^2}}$

$$\left(\cos^{-1} u\right)' = -\frac{u'}{\sqrt{1-u^2}}$$

$$\left(\tan^{-1} u\right)' = \frac{u'}{1+u^2}$$

$$\left(\cot^{-1} u\right)' = -\frac{u'}{1+u^2}$$

ຕົວຢ່າງ 2: ຈົ່ງຊອກຫາຜົນຕໍ່ລາວຂອງຕໍ່ລາວຕໍ່ໄປນີ້:

1) $y = \dfrac{1}{\sin^{-1} x}$

$$y' = -\frac{1}{(\sin^{-1} x)^2 \sqrt{1-x^2}}$$

2) $f(x) = x \tan^{-1} \sqrt{x}$

$$f'(x) = \tan^{-1} \sqrt{x} + x \cdot \frac{\left(\sqrt{x}\right)'}{1+x}$$

$$= \tan^{-1} \sqrt{x} + \frac{\sqrt{x}}{2(1+x)}$$

ຄຳຖາມ 4: ຈົ່ງຊອກຫາຜົນຕໍ່ລາວຂອງຕໍ່ລາວຕໍ່ໄປນີ້:

1) $y = \dfrac{1}{\cos^{-1} x}$

2) $f(x) = \tan^{-1} x^2$

3. ສະຫຼຸບ

$$(\sin x)' = \cos x$$

$$(\cos x)' = -\sin x$$

$$(\tan x)' = \frac{1}{\cos^2 x}$$

$$(\cot x)' = -\frac{1}{\sin^2 x}$$

$$\left(\sin^{-1} x\right)' = \frac{1}{\sqrt{1-x^2}}$$

$$\left(\cos^{-1} x\right)' = -\frac{1}{\sqrt{1-x^2}}$$

$$\left(\tan^{-1} x\right)' = \frac{1}{1+x^2}$$

$$\left(\cot^{-1} x\right)' = -\frac{1}{1+x^2}$$

ເມື່ອ u ເປັນຕໍ່ລາວຕາມ x ເຮົາໄດ້

$$(\sin u)' = u' \cos u$$

$$(\cos u)' = -u' \sin u$$

$$(\tan u)' = \frac{u'}{\cos^2 u}$$

$$(\cot u)' = -\frac{u'}{\sin^2 u}$$

$$\left(\sin^{-1} u\right)' = \frac{u'}{\sqrt{1-u^2}}$$

$$\left(\cos^{-1} u\right)' = -\frac{u'}{\sqrt{1-u^2}}$$

$$\left(\tan^{-1} u\right)' = \frac{u'}{1+u^2}$$

$$\left(\cot^{-1} u\right)' = -\frac{u'}{1+u^2}$$

ບົດຝຶກຫັດ

ຈົ່ງຄິດໄລ່ຜົນຕໍາລາ ຂອງຕໍາລາຕໍ່ໄປນີ້:

1) $f(x) = 2\sin x - 3\cos x$

2) $f(x) = \tan^2 3x$

3) $f(x) = \sin^3 x - \cos^3 x$

4) $f(x) = \ln(2 + \cos x)$

5) $f(x) = \tan(1 + x^4)$

6) $f(x) = \ln(1 + \cot^2 x)$

7) $f(x) = \dfrac{1}{\tan(2x + 1)}$

8) $y = x^2(\sin^{-1} x)^3$

9) $y = \tan^{-1}\sqrt{\dfrac{1-x}{1+x}}$

10) $y = \sin^{-1}(x^2 \ln x)$